25900317 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BCA (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Core Course (CC) - MG3CCRBCA203 - DESIGN AND ANALYSIS OF ALGORITHMS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. Given the code, give the space complexity (in terms of auxiliary space): [An] / [CO1]
   ```
   function f(n):
   if n == 0:return 1
   temp = array of size n
   return f(n-1) + temp
   ```
2. Define Recursion. [K] / [CO1]
3. What is the advantage of Quick Sort over Merge Sort? [U] / [CO2]
4. Discuss the worst-case complexity of binary search. [U] / [CO2]
5. Why does Strassen's algorithm use addition and subtraction operations along with multiplication? [U] / [CO2]
6. Explain how the Divide and Conquer technique is applied to find the maximum and minimum values in a list of elements. [U] / [CO2]
7. Define biconnected components. [K] / [CO3]

8. Define principle of optimality [K] / [CO3]

9. List any two characteristics of BFS. [K] / [CO3]

10. Define Backtracking. [K] / [CO3]

[2x10 = 20]

Part B

Short Answer Questions

Answer any **3** questions

Each question carries **5** marks

11. List out the advantages of using algorithm. [U] / [CO1]

12. Explain the divide and conquer general method? [U] / [CO2]

13. List the major steps in designing an algorithm using the greedy strategy. [K] / [CO2]

14. Write the step-by-step procedure of Dijkstra's Algorithm. [U] / [CO2]

15. Explain the Sum of Subsets problem with an algorithm. Solve it using the data set S = {3, 5, 6, 7} with d = 15. [An] / [CO3]

[5x3 = 15]

Part C

Essay Questions

Answer any **1** questions

Each question carries **15** marks

16. Discuss in detail the working mechanism, data structures used, and efficiency of Prim's algorithm. Illustrate with a suitable example [An] / [CO2]

17. Apply Hamiltonian cycle algorithm to find cycles in a 6-vertex complete graph. [A] / [CO3]

[15x1 = 15]